On Graph Reconstruction via Empirical Risk Minimization: Fast Learning Rates and Scalability, a Review

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Main Finding

Graph Reconstruction Achieves a Fast Learning Rate

- Classical VC theory gives

$$R(\hat{g}_n) - R^* = O\left(\frac{1}{\sqrt{n}}\right).$$

- Papa et al. (2016) show instead

$$R(\hat{g}_n) - R^* = O\left(\frac{\log n}{n}\right).$$

Scalability Note (Not Discussed Today)

- Papa et al. also propose an MCMC algorithm for large graphs.
- We omit this part, as it is not relevant to learning theory.

Graph Reconstruction Setup

- We observe a graph $G = (V, E)$ with

$$V = \{1, \ldots, n\}, \quad E = \{e_{ij} : 1 \leq i < j \leq n\}.$$

- Node features $X_i \overset{\text{i.i.d.}}{\sim} \mu$.
- Edges are undirected: $e_{ij} = e_{ji}$.
- Edge formation model:

$$\eta(x_i, x_j) = \mathbb{P}(e_{ij} = 1 \mid X_i, X_j).$$

- **Goal**: learn reconstruction rule

$$g : \mathcal{X}^2 \to \{0, 1\}$$

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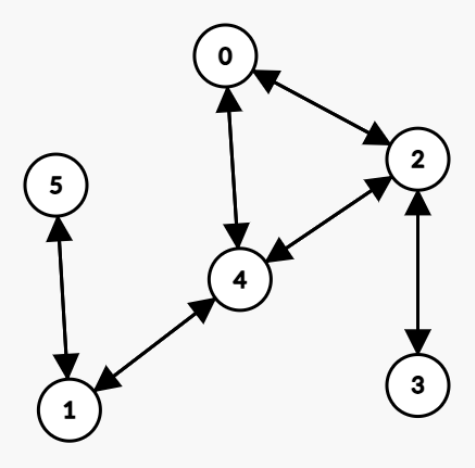

Undirected graph (6 nodes)

Empirical Risk

Minimizer (estimator): $\hat{g}_n = \arg\min_{g \in \mathcal{G}} \hat{\mathcal{R}}_n(g)$

Risk:

$$\hat{\mathcal{R}}_n(g) = \frac{2}{n(n-1)} \sum_{i<j} \ell(g(X_i, X_j), e_{ij})$$

U-Statistics: Crash Course

Definition (Order m)

A *U-statistic of order m* is

$$U_n = \frac{1}{\binom{n}{m}} \sum_{1 \le i_1 < \cdots < i_m \le n} h(X_{i_1}, \ldots, X_{i_m}),$$

where $h : \mathcal{X}^m \to \mathbb{R}$ is a symmetric kernel.

- U-statistics estimate quantities of the form

$$\theta = \mathbb{E}[h(X_1, \ldots, X_m)].$$

- They are **unbiased**: $\mathbb{E}[U_n] = \theta$.
- Many pairwise learning problems naturally produce U-statistics.

U-Statistics of Order 2

A degree-2 U-statistic has the form

$$U_n = \frac{1}{\binom{n}{2}} \sum_{i<j} h(X_i, X_j),$$

with symmetric kernel

$$h(x_1, x_2) = h(x_2, x_1).$$

Why order 2 matters here

Graph reconstruction risk is exactly this:

$$\widehat{R}_n(g) = \frac{1}{n(n-1)} \sum_{i \neq j} \ell(g(X_i, X_j), e_{ij}),$$

which is a pairwise average over all unordered node pairs.

Roadmap of the Result

Three Building Blocks

1. **Biau & Bleakley (2006)** Graph reconstruction risk is a *degree-2 U-statistic*.
2. **Hoeffding (1941)** Decomposition of any order-2 U-statistic into:

$$h = \theta + h_1 + h_1 + h_2.$$

3. **Clémençon et al. (2008)** Learning rates for U-processes via

$$\text{i.i.d. term} + \text{degenerate term}.$$

Papa's New Idea

- Center the loss at the Bayes rule $\Rightarrow$ variance of the i.i.d. term collapses.
- What remains is Clémençon's *degenerate* term.

Conclusion

$$R(\hat{g}_n) - R^* = O\left(\frac{\log n}{n}\right).$$

Biau and Bleakley (2006): Baseline Rate

Two key contributions:

1. **Graph reconstruction is a pairwise ERM problem.**

$$\widehat{R}_n(g) = \frac{1}{n(n-1)} \sum_{i \neq j} \ell(g(X_i, X_j), e_{ij}) \quad \Rightarrow \quad \text{order-2 U-statistic.}$$

2. **The reconstruction class $\mathcal{G}$ has finite VC dimension** (proved by Biau Bleakley).

Consequence: Clémençon et al. (2008) provide uniform deviation bounds for all pairwise ERM problems:

$$\sup_{g \in \mathcal{G}} |U_n(g) - R(g)| = O_{\mathbb{P}}\left(\sqrt{\frac{V}{n}} + \frac{V}{n}\right).$$

Therefore:

$$R(\hat{g}_n) - R^* = O\left(\frac{1}{\sqrt{n}}\right).$$

Empirical Risk is a U-Statistic of Order 2

- For any rule g, the empirical reconstruction risk is

$$\widehat{R}_n(g) = \frac{1}{n(n-1)} \sum_{i \neq j} \ell\big(g(X_i, X_j), e_{ij}\big),$$

  a symmetric average over all unordered pairs.
- Therefore it has the form of a **degree-2 U-statistic**:

$$U_n(g) = \frac{1}{\binom{n}{2}} \sum_{i<j} h_g(X_i, X_j, e_{ij}).$$

- **Key implication:** Hoeffding's decomposition applies.

What this gives us

Every order-2 U-statistic decomposes into

$$U_n(g) - R(g) = \frac{2}{n} \sum_{i=1}^{n} h_{1,g}(X_i) + U_{n,\mathrm{deg}}(g),$$

where the first term is an **i.i.d. process** and the second is a **degenerate U-statistic**.

Hoeffding Decomposition (Idea)

Goal: Decompose any order-2 U-statistic into:

$$\text{i.i.d. linear term} \;+\; \text{degenerate term.}$$

- Start with a symmetric kernel $h(x_1, x_2)$.
- Hoeffding writes it as:

$$h = \theta + h_1(x_1) + h_1(x_2) + h_2(x_1, x_2),$$

  where:
  - θ: grand mean,
  - h_1: first-order projection (i.i.d. part),
  - h_2: degenerate remainder (has mean 0 when one variable is fixed).

Purpose: Separate the "easy" i.i.d. variation from the "hard" dependent remainder.

Hoeffding's Decomposition (Details)

Given a symmetric kernel $h(x_1, x_2)$:

1. Grand mean:
$$\theta = \mathbb{E}[h(X_1, X_2)].$$
2. First-order projection:
$$h_1(x) = \mathbb{E}[h(x, X_2)] - \theta.$$
These terms are i.i.d. when evaluated at X_i.
3. Second-order projection (degenerate):
$$h_2(x_1, x_2) = h(x_1, x_2) - \theta - h_1(x_1) - h_1(x_2),$$
and satisfies
$$\mathbb{E}[h_2(X_1, X_2) \mid X_1] = 0.$$

Hence:
$$h(x_1, x_2) = \theta + h_1(x_1) + h_1(x_2) + h_2(x_1, x_2).$$

Applying Hoeffding to the U-Statistic

For an order-2 U-statistic

$$U_n(g) = \frac{1}{\binom{n}{2}} \sum_{i<j} h(X_i, X_j),$$

substitute the decomposition:

$$U_n(g) - R(g) = \underbrace{\frac{2}{n} \sum_{i=1}^{n} h_1(X_i)}_{\text{i.i.d. linear term}} + \underbrace{\frac{1}{\binom{n}{2}} \sum_{i<j} h_2(X_i, X_j)}_{\text{degenerate term}}.$$

- The linear term behaves like a standard empirical process.
- The degenerate term averages dependent pairs but has zero conditional mean.

This decomposition is the backbone of Clémençon and Papa's analyses.

Clémençon et al. (2008): Goal

Goal: Uniform control of pairwise ERM

$$\sup_{g \in \mathcal{G}} \left| U_n(g) - R(g) \right|$$

Using Hoeffding's decomposition:

$$U_n(g) - R(g) = \underbrace{\frac{2}{n} \sum_{i=1}^{n} h_{1,g}(X_i)}_{\text{i.i.d. linear term}} + \underbrace{U_{n,\text{deg}}(g)}_{\text{degenerate term}} .$$

- The problem splits into two parts:
  - **i.i.d. term:** standard empirical process.
  - **degenerate term:** dependent U-statistic, much harder.

Clémençon: Bounding the i.i.d. Term

Linear part:

$$\frac{2}{n}\sum_{i=1}^{n} h_{1,g}(X_i), \qquad X_i \overset{\text{i.i.d.}}{\sim} \mu.$$

- This is a standard empirical process indexed by $g \in \mathcal{G}$.
- For a VC class of size V,

$$\sup_{g\in\mathcal{G}} \left| \frac{2}{n}\sum_{i=1}^{n} h_{1,g}(X_i) \right| = O_{\mathbb{P}}\left(\sqrt{\frac{V}{n}}\right).$$

Conclusion: The i.i.d. part behaves just like ordinary classification.

Clémençon: The Hard Part — Degenerate Term

Degenerate term:

$$U_{n,\mathrm{deg}}(g) = \frac{1}{\binom{n}{2}} \sum_{i<j} h_{2,g}(X_i, X_j).$$

- Terms are **dependent**: each X_i appears in $n-1$ summands.
- Clémençon's insight: A **random permutation** creates $m = \lfloor n/2 \rfloor$ *disjoint* pairs that behave like i.i.d. pairs.

Permutation-based approximation:

$$\tilde{U}_n = \frac{1}{n!} \sum_{\sigma \in S_n} \frac{1}{m} \sum_{r=1}^{m} h_2(X_{\sigma(r)}, X_{\sigma(r+m)}).$$

Key fact:

$$\mathbb{E}[U_{n,\mathrm{deg}}(g) \mid X] = \mathbb{E}[\tilde{U}_n \mid X].$$

Therefore:

$$\sup_{g \in \mathcal{G}} |\tilde{U}_n| = O_{\mathbb{P}}\left(\sqrt{\frac{V}{n}} + \frac{V}{n}\right).$$

Papa's Starting Point

From Clémençon et al. (2008):

$$U_n(g) - R(g) = \underbrace{\frac{2}{n}\sum_{i=1}^{n} h_{1,g}(X_i)}_{\text{i.i.d. linear term}} + \underbrace{U_{n,\deg}(g)}_{\text{degenerate term}} .$$

- Clémençon shows the **degenerate term** already has a fast rate:

$$U_{n,\deg}(g) = O_{\mathbb{P}}\left(\frac{\log n}{n}\right).$$

- The bottleneck is the **i.i.d. linear term**:

$$\frac{2}{n}\sum_{i=1}^{n} h_{1,g}(X_i) \text{ typically fluctuates at } O\left(\sqrt{\frac{1}{n}}\right).$$

Papa's idea: Make the i.i.d. term *smaller* by analyzing its variance under Bayes-centering.

Papa's Key Step: Bayes-Centered Loss

Define the Bayes rule:

$$g^*(x_1, x_2) = \mathbf{1}\{\eta(x_1, x_2) > 1/2\}.$$

Define the **Bayes-centered excess loss**:

$$q_g = \ell(g, e) - \ell(g^*, e).$$

Then:

$$\mathbb{E}[q_g] = R(g) - R(g^*) = \Lambda(g) \quad \text{(excess risk)}.$$

Papa studies the Hoeffding projection of q_g:

$$h_{1,g}(x_1) = \mathbb{E}[q_g(x_1, X_2, e_{12}) \mid X_1 = x_1].$$

Crucial lemma:

$$\mathrm{Var}(h_{1,g}(X_1)) \;\; \leq \;\; \Lambda(g).$$

Interpretation: The i.i.d. term becomes *automatically small* whenever g is close to Bayes.

Why Bayes-Centering Gives a Fast Rate

The i.i.d. Hoeffding term:

$$\frac{2}{n}\sum_{i=1}^{n} h_{1,g}(X_i)$$

now satisfies a **Bernstein-type condition**:

$$\mathrm{Var}(h_{1,g}) \leq \Lambda(g).$$

This implies:

$$\sup_{g\in\mathcal{G}}\left|\frac{2}{n}\sum_{i=1}^{n} h_{1,g}(X_i)\right| = O_{\mathbb{P}}\left(\sqrt{\frac{\Lambda(g)\log n}{n}}\right).$$

- When g is nearly optimal, $\Lambda(g)$ is small.
- The i.i.d. term becomes **negligible** compared to the degenerate term.

Therefore:

$$R(\hat{g}_n) - R^* = O\left(\frac{\log n}{n}\right).$$

Conclusion: Three Key Ideas Behind Fast Rates

1. **Hoeffding U-Statistic Decomposition**
   - Pairwise ERM in graph reconstruction is a **U-statistic of order 2**.
   - Hoeffding decomposes it into:

$$U_n - R = \underbrace{\text{i.i.d. linear term}}_{\text{slow, } O(1/\sqrt{n})} + \underbrace{\text{degenerate U-term}}_{\text{fast, } O(1/n)}.$$

2. **Clémençon et al. (2008): Controlling the U-statistic**
   - Showed that dependent graph pairs can be analyzed as if we averaged over i.i.d. pairs via symmetrization and decoupling.
   - Gave tight bounds for the degenerate component, already yielding fast decay.
3. **Papa et al. (2016): Variance Control of the i.i.d. Term**
   - After Bayes-centering,

$$\mathrm{Var}\big(h_{1,g}(X)\big) \le \Lambda(g) = R(g) - R^*.$$

   - The "slow" i.i.d. term shrinks with the excess risk, causing the fast degenerate term to dominate.
   - Result:

$$R(\hat{g}_n) - R^* = O\left(\frac{\log n}{n}\right).$$

Thank You

Appendix A: Hoeffding vs. Bernstein Inequalities

Hoeffding's Inequality (Variance-Blind)

$$\mathbb{P}\left(\left|\frac{1}{n}\sum_{i=1}^{n} Z_i - \mathbb{E}Z_i\right| > t\right) \leq 2\exp\left(-\frac{2nt^2}{(b-a)^2}\right)$$

- Only uses the **range** of Z_i (boundedness).
- Gives fluctuations of order:

$$O\left(\sqrt{\frac{1}{n}}\right).$$

- Same rate even if the variance is tiny.

Bernstein's Inequality (Variance-Adaptive)

$$\mathbb{P}\left(\left|\frac{1}{n}\sum_{i=1}^{n} Z_i - \mathbb{E}Z_i\right| > t\right) \leq 2\exp\left(-\frac{nt^2}{2\sigma^2 + \frac{2}{3}t(b-a)}\right)$$

- Adapts to the **actual variance**: $\sigma^2 = \mathrm{Var}(Z_i)$.
- Fluctuations scale like

$$O\left(\sqrt{\frac{\sigma^2}{n}}\right).$$

Appendix B: Expectation of the Permutation Estimator

Goal. Show that, conditionally on X_1^n,

$$\mathbb{E}[\tilde{U}_n \mid X] = \mathbb{E}[U_{n,\deg}(g) \mid X].$$

Permutation estimator:

$$\tilde{U}_n = \frac{1}{n!} \sum_{\sigma \in S_n} \frac{1}{m} \sum_{r=1}^{m} h_2\left(X_{\sigma(r)}, X_{\sigma(r+m)}\right), \qquad m = \lfloor n/2 \rfloor.$$

Idea. A random permutation produces m disjoint pairs that "simulate" i.i.d. pairs. We compute the expectation explicitly using counting variables.

Appendix B: Counting Argument

Define the counting variable

$$N_{i,j}(\sigma) = \sum_{r=1}^{m} \mathbf{1}\{\{\sigma(r), \sigma(r+m)\} = \{i,j\}\}\,.$$

Then we can rewrite:

$$\tilde{U}_n = \frac{1}{n!} \sum_{\sigma \in S_n} \frac{1}{m} \sum_{i<j} N_{i,j}(\sigma)\, h_2(X_i, X_j).$$

Conditional expectation (conditioning on X):

$$\mathbb{E}[\tilde{U}_n \mid X] = \frac{1}{m} \sum_{i<j} h_2(X_i, X_j)\, \mathbb{E}_\sigma[N_{i,j}(\sigma)].$$

Appendix B: Expected Counts and Result

Compute $\mathbb{E}_\sigma[N_{i,j}(\sigma)]$.
For any $r \leq m$,

$$\mathbb{P}(\{\sigma(r), \sigma(r+m)\} = \{i,j\}) = \frac{1}{\binom{n}{2}}$$

because a random permutation is uniformly distributed over all unordered pairs.
Thus:

$$\mathbb{E}[N_{i,j}(\sigma)] = m \cdot \frac{1}{\binom{n}{2}}.$$

Plug in.

$$\mathbb{E}[\tilde{U}_n \mid X] = \frac{1}{\binom{n}{2}} \sum_{i<j} h_2(X_i, X_j) = U_{n,\mathrm{deg}}(g).$$

Conclusion. The permutation estimator has the *same conditional expectation* as the degenerate U-term. Thus, the degenerate part behaves like an average over m almost i.i.d. pairs.

Appendix C: Lemma 2 (Variance Control)

Lemma 2

For any reconstruction rule g,

$$\mathrm{Var}(\mathbb{E}[q_g(X_1, X_2, e_{12}) \mid X_1]) \;\leq\; \Lambda(g),$$

where $\Lambda(g) = R(g) - R^* = \mathbb{E}[q_g]$ is the excess risk.

Definitions to remember:

- $q_g = H_g - H_{g^*}$ = excess 0–1 loss vs. Bayes rule.
- $g^*(x_1, x_2) = \mathbf{1}\{\eta(x_1, x_2) > 1/2\}$.
- $\mathbb{E}[q_g \mid X_1]$ = expected excess loss when one endpoint is fixed.

Intuition: The conditional excess loss is nonzero only when g disagrees with g^*, and is weighted by the margin $|2\eta - 1|$. This structure forces its variance to be $\leq$ the mean excess loss.

Appendix C: Key Conditional Expectations

Setup:

$$\eta(x_1, x_2) = \mathbb{P}(e = 1 \mid x_1, x_2), \qquad H_g = \mathbf{1}\{g \neq e\}.$$

Expected error of any rule g

$$\mathbb{E}[H_g \mid X_1, X_2] = \begin{cases} \eta(X_1, X_2), & g = 0, \\ 1 - \eta(X_1, X_2), & g = 1. \end{cases}$$

Expected error of Bayes rule

$$\mathbb{E}[H_{g^*} \mid X_1, X_2] = \min(\eta, 1 - \eta) = \tfrac{1}{2} - \left|\eta - \tfrac{1}{2}\right|.$$

Key identity for $q_g = H_g - H_{g^*}$

$$\mathbb{E}[q_g \mid X_1, X_2] = |2\eta(X_1, X_2) - 1|\, \mathbf{1}\{g(X_1, X_2) \neq g^*(X_1, X_2)\}.$$

Appendix C: Why the Variance is Bounded by Excess Risk

Let

$$Z(X_1) := \mathbb{E}[q_g(X_1, X_2, e_{12}) \mid X_1].$$

From the previous slide:

$$Z(X_1) = \mathbb{E}\Big[\, |2\eta - 1|\, \mathbf{1}\{g \neq g^*\} \mid X_1 \Big].$$

Bounding the variance:

$$\mathrm{Var}(Z) \;\leq\; \mathbb{E}[Z^2] \;\leq\; \mathbb{E}[Z] \qquad (\text{since } 0 \leq Z \leq 1).$$

But

$$\mathbb{E}[Z] = \mathbb{E}[q_g] = \Lambda(g),$$

the excess risk.

Conclusion

$$\mathrm{Var}(\mathbb{E}[q_g \mid X_1]) \;\leq\; \Lambda(g).$$

This "variance collapse" is the key ingredient enabling Papa's fast rate.